

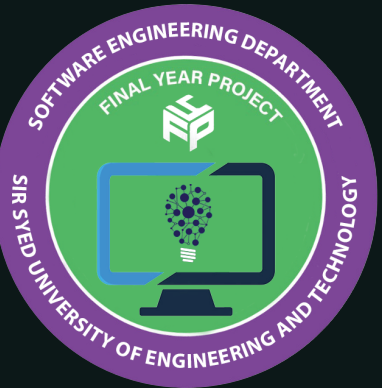


Izma Wara Khan
(2022F-BSE-184)
DilAfroze
(2022F-BSE-193)

PROJECT 40: AI Legal Assistant

PROJECT SUPERVISOR NAME: Engr. Nida Khalil/Engr. Priha Bhatti

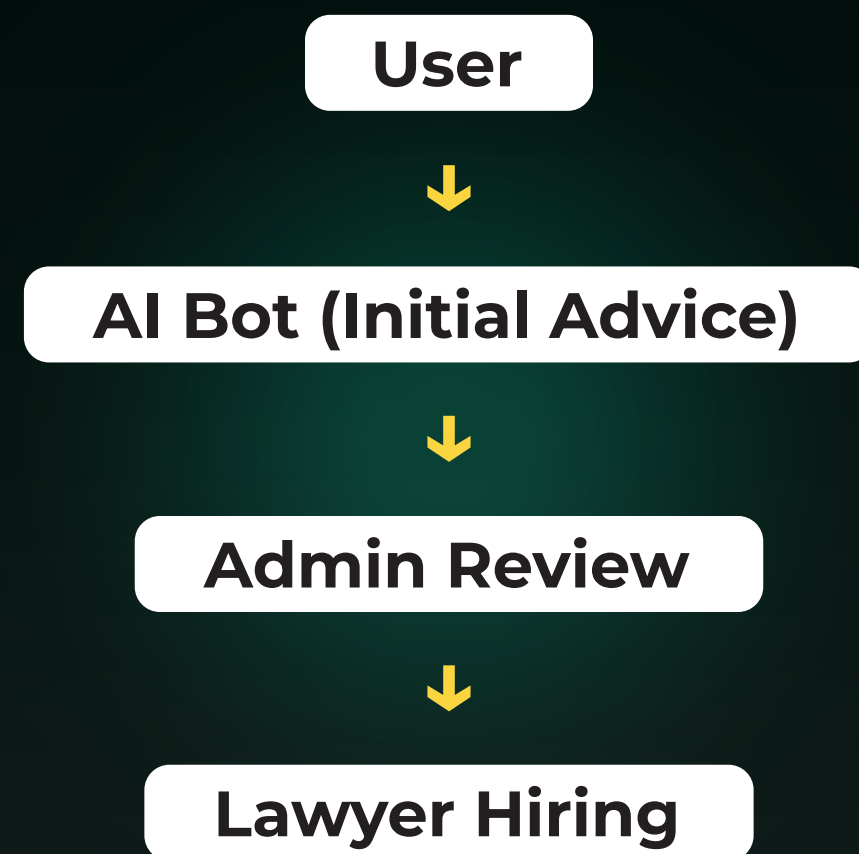
M. Bilal
(2022F-BSE-190)



PROJECT SCOPE

- **Target Users:**
General public, law students & legal professionals in Pakistan.
- **Legal Areas:**
Family, Criminal, Banking, Labour & International Law.
- **Focus:**
Rural Sindh & safe, anonymous access for female users.

SYSTEM MODEL



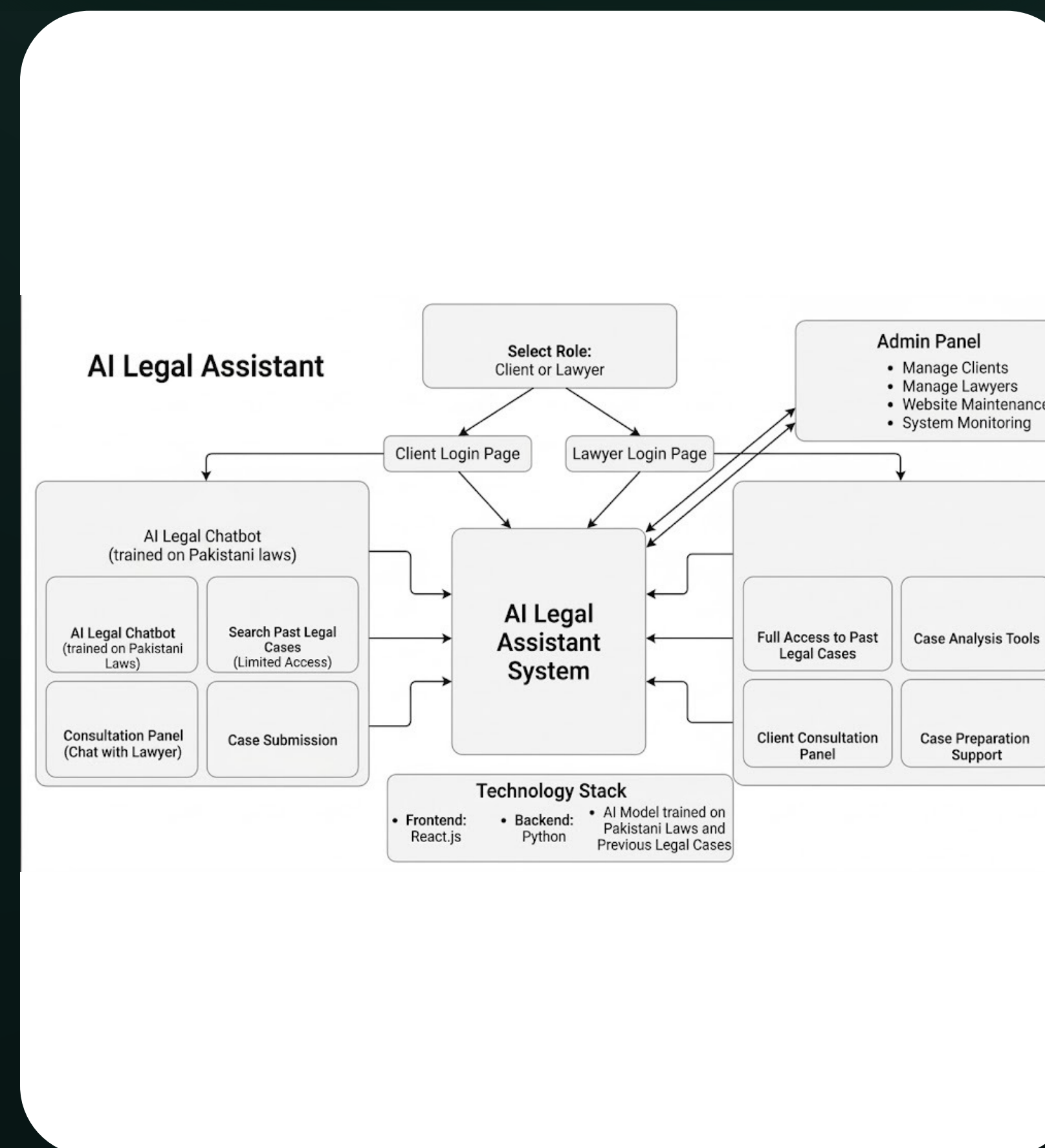
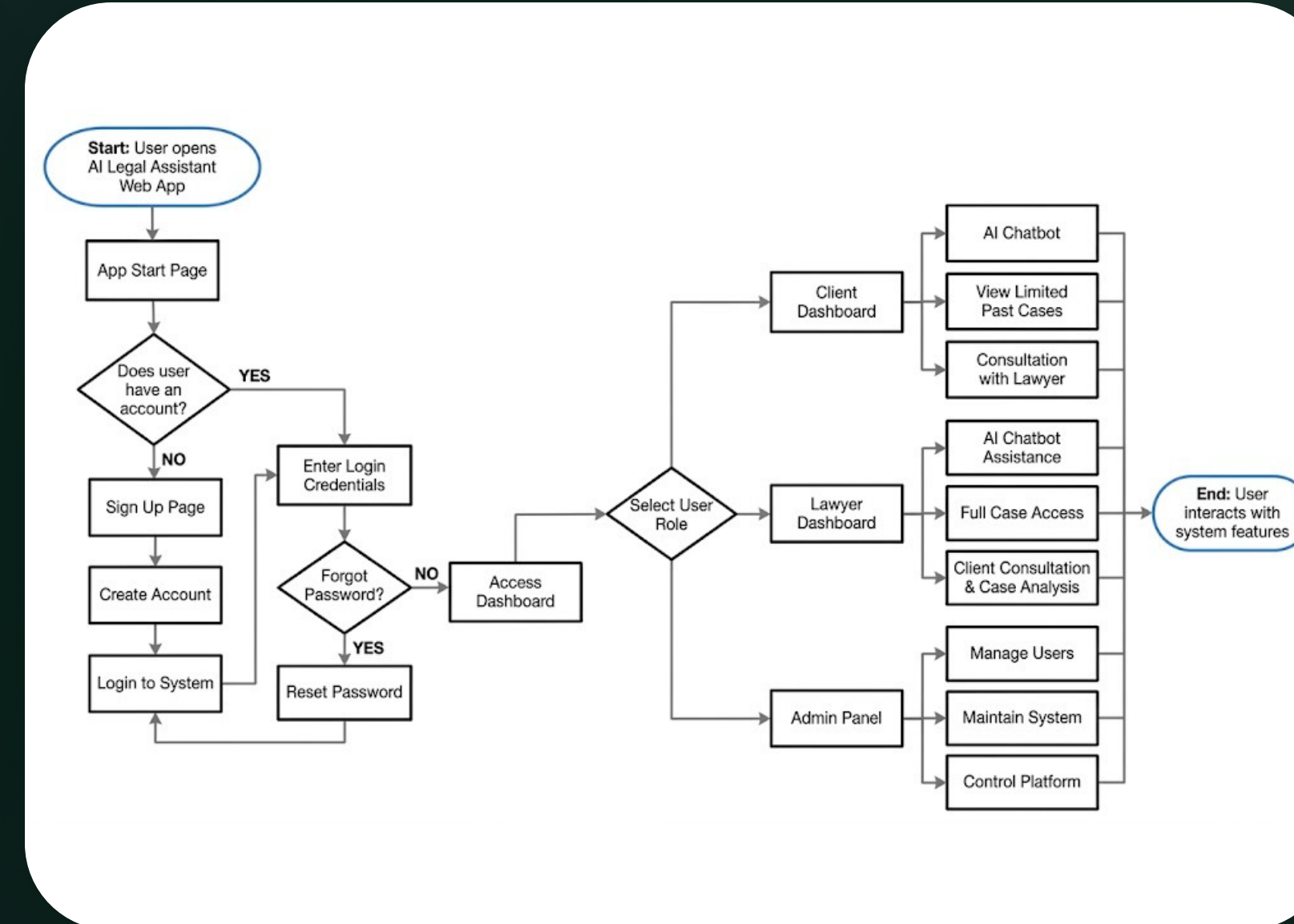
RAG ensures answers are based on authentic legal statutes.

KEY FEATURES

- Secure online payments for consultations, legal fees, and subscriptions.
- AI chatbot delivering relevant legal guidance.
- Lawyer search with live chat support.
- End-to-end encrypted communication.
- Dedicated case dashboard with history and outcomes.
- Separate portals for Client, Lawyer, and Admin.

KEY FEATURES

The platform runs on a full stack AI architecture built for Pakistan's legal system, using a 3 panel setup for Client, Lawyer, and Admin to streamline service delivery. It processes text and voice queries through NLP to detect language and map them to the right legal domain. Responses are generated via a RAG engine using LangChain and Llama 3.3, pulling from a verified Pakistani legal database. All user data, sessions, and case records are securely stored in MongoDB or SQL Server for persistence and efficient tracking.



PROJECT GOALS

- Streamline case tracking and management for clients and lawyers.
- Affordable, accessible legal consultation.
- Smart case history panel for better decision-making.
- Strong lawyer client connectivity.
- High level data and session security.
- Clean, intuitive interface for seamless navigation.
- Promote legal awareness.

TECHNOLOGY USED

- **Frontend:** React JS / Flutter
- **Backend:** Python (FastAPI), Node.js
- **AI:** Groq (Llama 3.3), LangChain, OpenAI API
- **Database:** SQL Server / MongoDB